

AUTOMATIC **S**PEECH **R**ECOGNITION

12 Feb 2026

&

TEXT
To
SPEECH

Part 1:

ASR Fundamentals

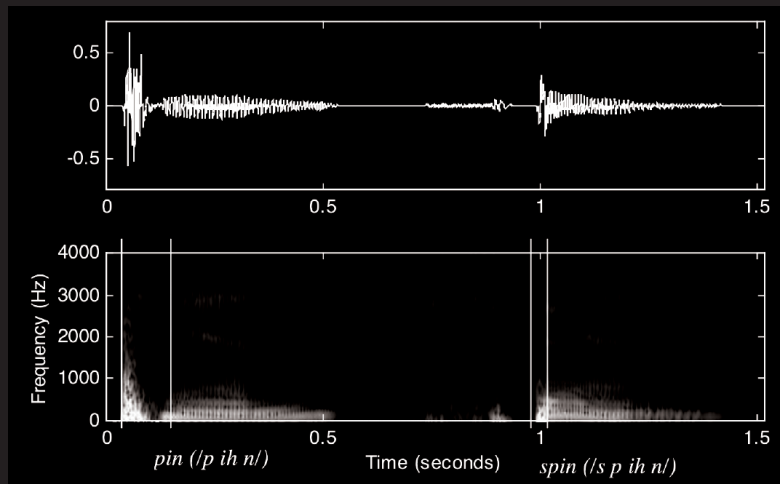
ASR types

- fixed commands
- grammars (e.g. date)
- continuous speech recognition (huge vocabulary)

What is ASR

- The aim is to decode the acoustic signal O into a word sequence $\hat{W}$ which is hopefully close to the original word sequence W .
- We need an **acoustic model**: our knowledge about acoustics, phonetics, microphone, environment, speakers, dialects, etc.
- And a **language model** that knows about words (and non-words), how they co-occur and form sequences (utterances)

Challenge: context variability



Variability in speech

- context variability (pin vs spin),
coarticulation
- style variability
- speaker variability
- environment variability

Formal definition

Formal definition

- We can observe the set of parameters O (observations) for the acoustic signal.
- In the following equation acoustic model would be responsible for $P(O|W)$ and language model for $P(W)$

$$W = \arg \max_W P(W|O) = \arg \max_W \frac{P(O|W)P(W)}{P(O)} = \arg \max_W P(O|W)P(W)$$

But how do we calculate $P(O|W)$?

- Can we use words?

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- Can we use words? Yes, but then there will be a huge range of observations.
- Can we minimise the number of possible observations? Yes, by selecting some smaller phonetic units.

Why word is not a good unit:

- New task can have new words but we won't have any training data.
- There are too many words, and each of them has too many acoustic realisations.

Good unit is:

- **accurate**, in order to represent acoustic realisation in different contexts
- **trainable**, we should have enough data to estimate the parameters of the unit
- **generalisable**, so new words could be derived from our units

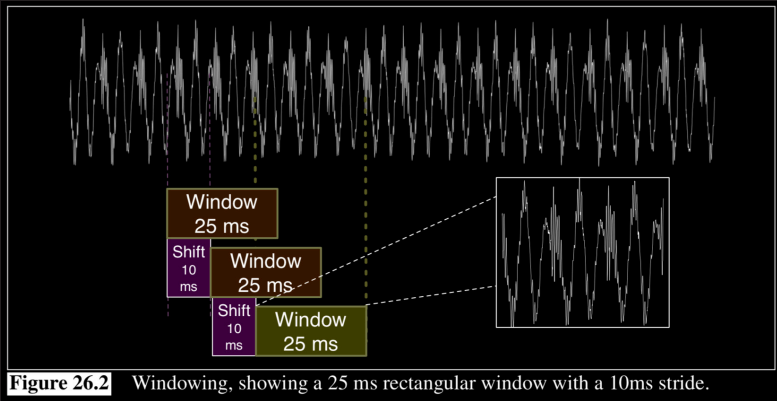
What about phonemes?

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- Just 50 phones in English, no problem to train. And they are vocabulary-independent.
- But phonemes are not produced independently, they are context-dependent, so they can over-generalise.
- For some languages we can use syllables (1200 in Chinese, 50 in Japanese) but for English (30k+) training is challenging.

Feature extraction

Feature extraction I: windowing



Hamming window

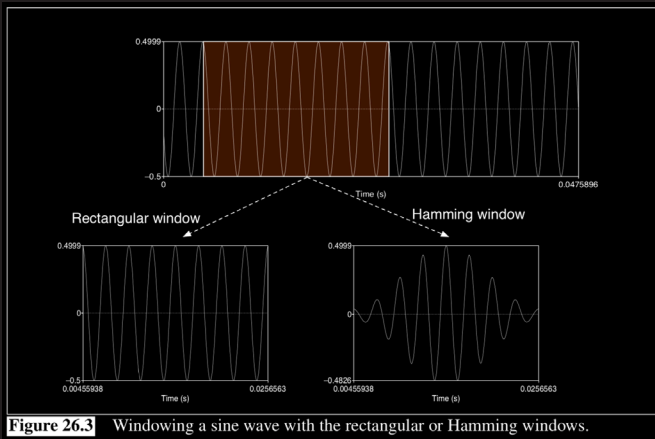


Figure 26.3 Windowing a sine wave with the rectangular or Hamming windows.

Discrete Fourier Transform (DFT)

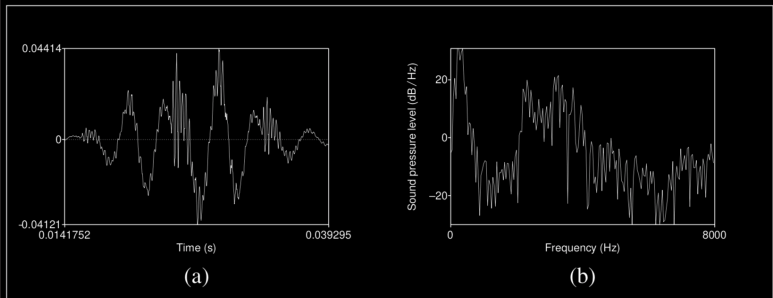


Figure 26.4 (a) A 25 ms Hamming-windowed portion of a signal from the vowel [iy] and (b) its spectrum computed by a DFT.

Mel filter bank

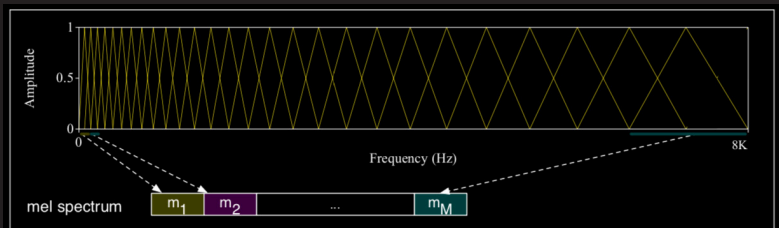
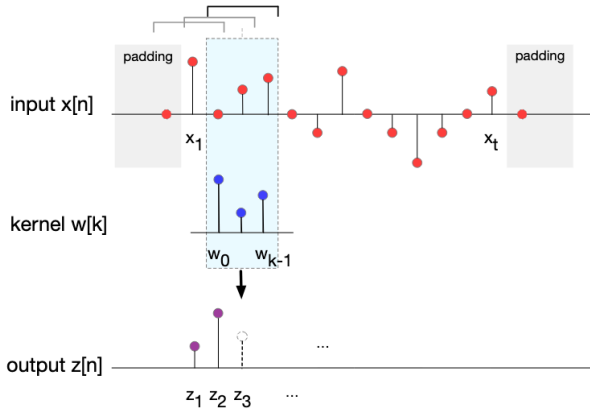


Figure 26.5 The mel filter bank (Davis and Mermelstein, 1980). Each triangular filter, spaced logarithmically along the mel scale, collects energy from a given frequency range.

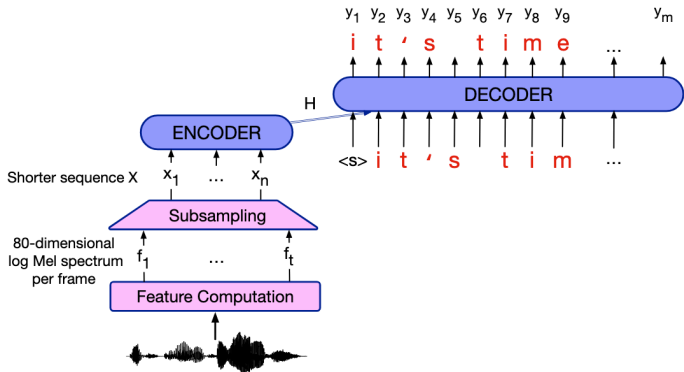
Convolutional Neural Network (CNN)



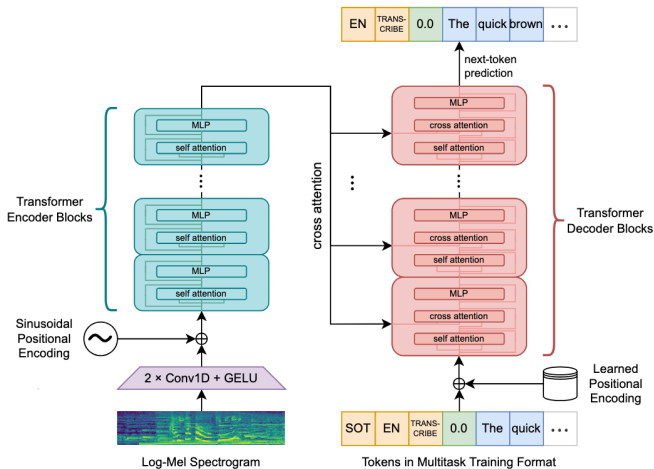
ASR architecture

- frames (width: 25ms, shift: 10ms), feature extraction, Hamming window
- Mel-frequency cepstral coefficients (MFCCs) are used as an input to individual Gaussian distribution for each phone
- encoder models
- decoding is integral path: the process which transforms the signal into word hypotheses.

Encoder-decoder architecture



Whisper (OpenAI)



Connectionist Temporal Classification (CTC)

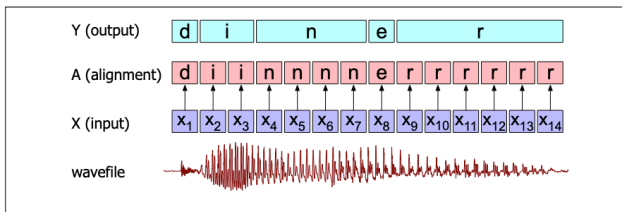


Figure 15.12 A naive algorithm for collapsing an alignment between input and letters.

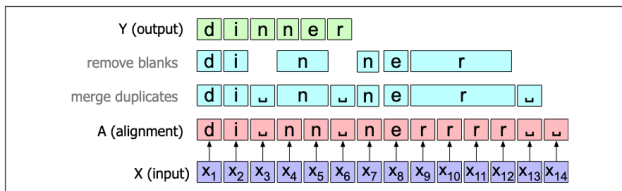
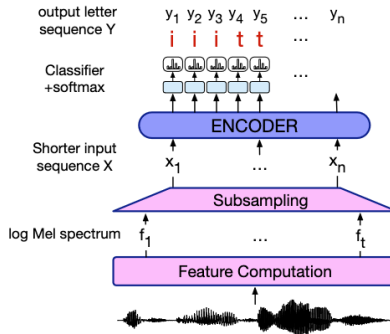


Figure 15.13 The CTC collapsing function B , showing the space blank character $\sqcup$; repeated (consecutive) characters in an alignment A are removed to form the output Y .

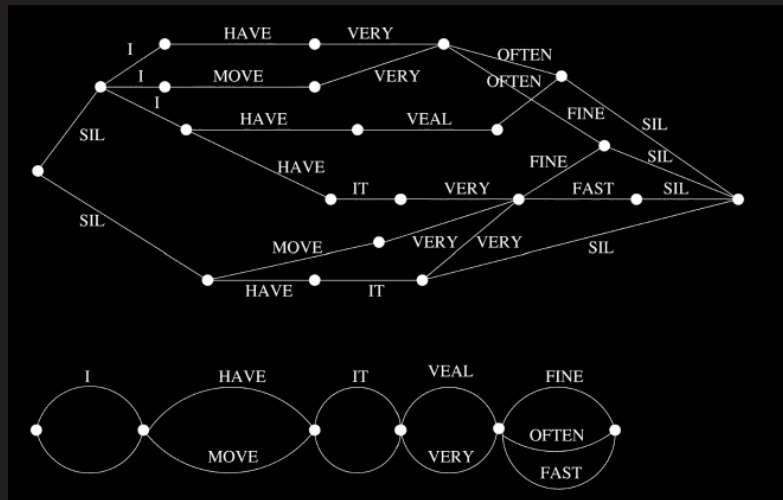
Connectionist Temporal Classification (CTC)



ASR decoding:

- Using Viterbi or other methods, multiple 'top' hypotheses can remain
- The possible outcomes can be stored in a word-confusion network (sausage) or a lattice.

Word lattice and confusion network



ASR evaluation

- Levenshtein distance/alignment
- Word-error rate (WER)

$$\text{WER} = (I + S + D) / N$$

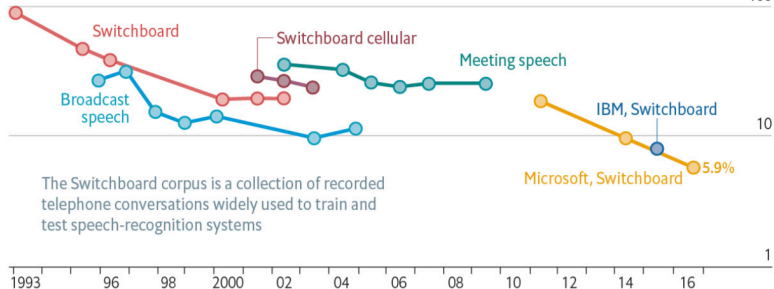
English Tasks	WER%
LibriSpeech audiobooks 960hour clean	1.4
LibriSpeech audiobooks 960hour other	2.6
Switchboard telephone conversations between strangers	5.8
CALLHOME telephone conversations between family	11
AMI meetings	11
Sociolinguistic interviews, CORAAL (AAE)	16.2
CHiME6 dinner parties with distant microphones	25.5
Sample tasks in other languages	WER%
Common Voice 15 Vietnamese	39.8
Common Voice 15 Swahili	51.2
FLEURS Bengali	50
Chinese (Mandarin) Tasks	CER%
AISHELL-1 Mandarin read speech corpus	3.9
HKUST Mandarin Chinese telephone conversations	18.5

Figure 15.1 Rough Word Error Rates (WER = % of words misrecognized) reported around 2023-4 for ASR on various American English and other language recognition tasks, and character error rates (CER) for two Chinese recognition tasks.

Loud and clear

Speech-recognition word-error rate, selected benchmarks, %

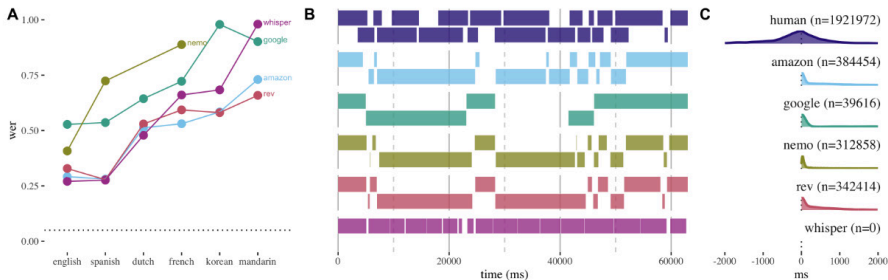
Log scale
100



The Switchboard corpus is a collection of recorded telephone conversations widely used to train and test speech-recognition systems

Sources: Microsoft; research papers

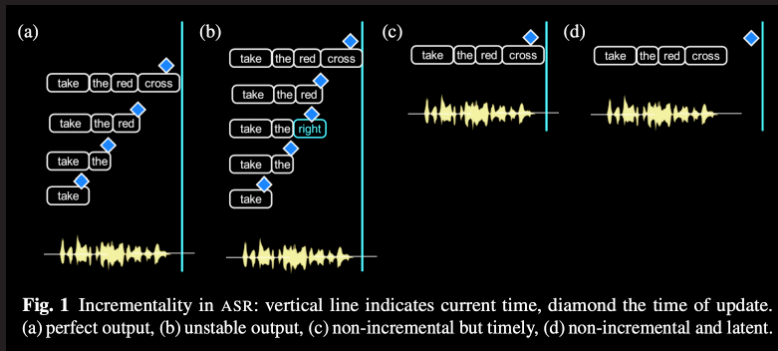
ASR and conversation



A Word error rates (WER) for five speech-to-text systems in six languages. **B** One minute of English conversation as annotated by human transcribers (top) and by five speech-to-text systems, showing that while most do some diarization, all underestimate the number of transitions and none represent overlapping turns (Whisper offers no diarization). **C** Speaker transitions and distribution of floor transfer offset times (all languages), showing that even ASR systems that support diarization do not represent overlapping annotations in their output.

Incremental ASR

Incrementality in ASR



Incrementalising ASR

- We need incremental ASR, which would require incrementalising all the internal processes.
- We need incremental LMs and AMs, in order to get the best sequences unit-by-unit.
- It is critical that ASR will provide timing for recognised words in a timely manner.

ASR should support disfluencies

“have the engine [take the oranges to Elmira, + { um, I mean, } take them to Corning] ” (Core and Schubert 1999)

- ‘um’ and ‘uh’ should be considered English words
- it is important to have access to “the oranges” in ASR output
- incremental disfluency detector (Hough and Purver, 2014)

Incremental evaluation I (Baumann et al., 2017)

Utterance-level Accuracy and Disfluency Suitability

- WER disfluency gain to determine how much of disfluent material is recovered

Incremental evaluation II

Timing: first occurrence (FO) and final decision (FD):

- FO is the time between the (true) beginning of a word and the first time it occurs in the output (regardless if it is afterwards changed)
- FD is the time between the (true) end of a word and the time when the recognizer decides on the word, without later revising it anymore.

Incremental evaluation III

Diachronic Evolution: how often consuming processors have to re-consider their output and for how long hypotheses are likely to still change.

- stability of the hypotheses. For words that are added and later revoked or substituted we measure the “survival time” and report aggregated plots of word survival rate (WSR) after a certain age.

Systems

- Off-the-shelf web-based ASR models have improved to a very good overall accuracy level. They can also be fine-tuned.
- Web Speech API and ponyfilling
- Training your ASR for your language/domain to improve accuracy can reap rewards (Sphinx, Kaldi).

Part 2:

TTS Fundamentals

TTS challenges

- coarticulation
- Text normalisation
- Homography
- Code-switching (and borrowed words)
- Morphology

Text normalisation

- abbreviations and acronyms: Dr., DC, NASA, COVID-19
- number formats, e.g. IBM 370, dates, times and currencies
- ~ Ü * " UPPER CASE :-)

Homograph normalisation

- Homograph variation can often be resolved on PoS (grammatical) category, e.g. object, bass, absent, -ate.
- But sometimes PoS does not help, e.g. read, kinda
- Variation of dialects
- Rate of speech (e.g. 'g' in recognise)

What is TTS

- text and phonetic analysis, grapheme-to-phoneme, morphology
- prosody generation/modification
- speech synthesis

Prosody

- speaking style (character and emotion)
- symbolic prosody (pauses, punctuation)
- accent and stress
- tone (e.g. yn- vs. wh-questions)

Speech synthesis

- Articulatory synthesis
- Rule-based formant synthesis
- Concatenative synthesis
- Statistical parametric synthesis
(generative synthesis)
- Deep learning-based synthesis

Concatenative synthesis (unit-selection)

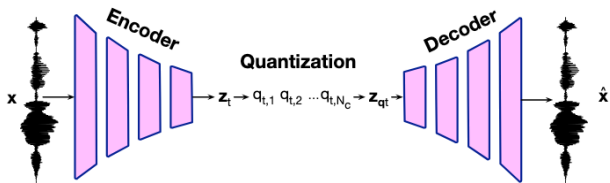
- Searches for the best sequence of (variably sized) units of speech in a large, annotated corpus of recordings, aiming to find a sequence that closely matches the target sequence.
- Sounds good if you fit the bits well. Also can be tuned.
- Can be really expensive to build: need professional speakers, many hours of stable recordings.

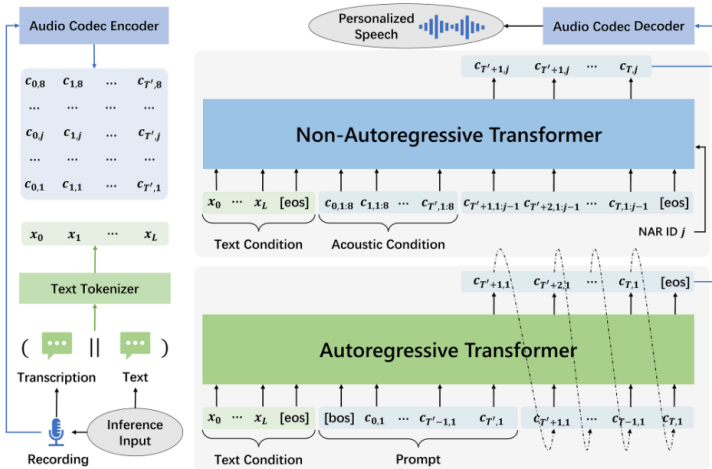
Statistical parametric speech synthesis

- Essentially, reproduces the speech signal.
- Break speech into several factors and represent them as numbers.
- Learns the model from data.
- From the model we can generate a spectrogram and then create a waveform.

Neural speech synthesis

- Essentially it is a type of statistical parametric speech synthesis
- text to features and then features to sound (neural vocoder)
- sequence2sequence models (e.g. RNN, GRU), Tacotron2
- e.g. LPCNet, WaveNet





Some advanced matters

- Simon King - Using Speech Synthesis to give Everyone their own Voice: <https://youtu.be/xzL-pxcpo-E>
- Incrementality: INPRO_iSS <https://www.youtube.com/watch?v=kwNvcUXfD7Y>

Evaluation

- Intelligibility tests, e.g. rhyme tests
- Quality tests: mean opinion scores (MOS) and preference test
- Functional tests
- Automated tests, mainly for letter-to-sound or prosody